Selection Sort Time Complexity:

$$C(n) = \sum_{i=0}^{n-2} \sum_{j=i+1}^{n-1} 1 = \sum_{i=0}^{n-2} (n-1-i+1)$$

$$= (n-1) + (n-2) + (n-3) + \dots + (1) = \frac{n(n-1)}{2} \in \Theta(n^2)$$

Stable Sorting Algorithm:

If we have two identical elements, a stable sorting algorithm always preserves their relative locations.

$1\ 5^{*}\ 3\ 2\ 5^{**}$ —after sorting→ $1\ 2\ 3\ 5^{**}\ 5^{*}$

if this is possible, then sorting alg. is NOT stable.

$1 \mid \check{5}^{*}\ 3\ 4\ 5^{**}\ 2$

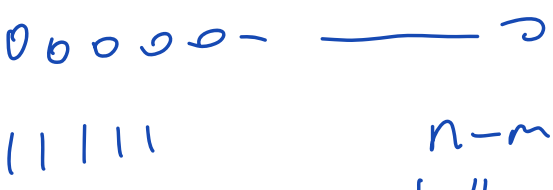
$1\ 2 \mid 3\ 4\ 5^{**}\ 5^{*}$

$1\ 2\ 3\ 4\ 5^{**}\ 5^{*}$ → Selection sort is not stable.

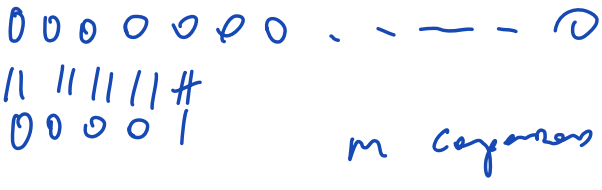
Brute-force string matching:

Best case:  - pattern is at the beginning of the text.

m comparisons

Worst case: 

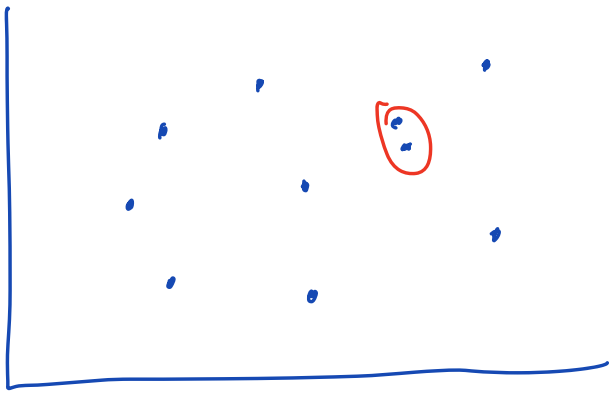
n-m+1 comparisons.
Is this a worst case?



m comparisons & (n-m+1) shift.

comparisons: $m(n-m+1)$ //

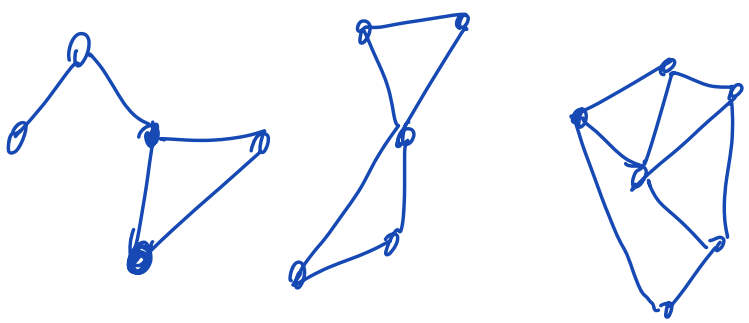
Closest Pair Problem

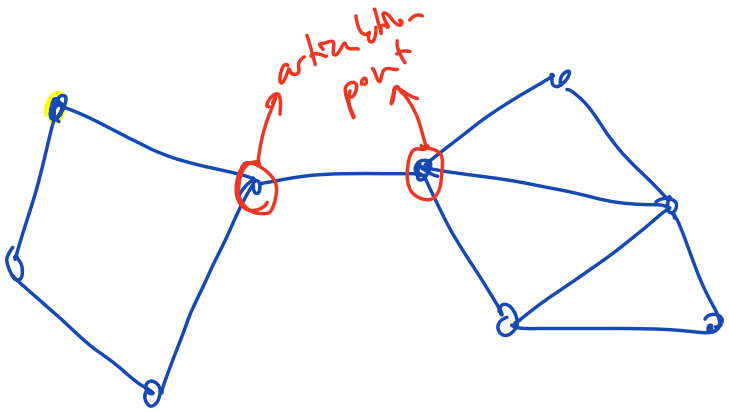


TSP by Exhaustive Search

Choose any two cities (except a) : b, c

Consider only the tours where b is before c




articulation point